

EC5.206 – Introduction to Statistical Signal Processing – Quiz

Date: 31st March, 2026

Instructor: Santosh Nannuru

Maximum marks: 50

Exam duration: 60 minutes

- a) Mention any additional assumptions you make that is not given in the question.
- b) Clearly show the steps used to arrive at the solutions.

1. [8 marks] Let $x[n]$ and $y[n]$ be two zero-mean random processes and $z[n] = x[n] + y[n]$.

- (a) If $x[n]$ and $y[n]$ have cross correlation $r_{xy}[n_1, n_2] = \sigma^2 \delta[n_1 - n_2]$, find the autocorrelation of $z[n]$ in terms of autocorrelation of $x[n]$ and $y[n]$.
- (b) If $x[n]$ and $y[n]$ are uncorrelated, find the autocorrelation of $z[n]$.

2. [12 marks] A random process $x[n]$ is given as

$$x[n] = \phi + \cos(\pi n + \phi)$$

where ϕ is a uniform random variable in $[0, 2\pi]$.

- (a) Sketch three realizations of this random process.
- (b) Find the mean and auto-correlation functions for this process.
- (c) Is this process wide sense stationary (WSS)?

3. [10 marks] A zero-mean white noise process $v[n]$ with unit variance is given as input to a causal LTI system with impulse response $h[n]$ given by $h[0] = 1$ and $h[1] = -\frac{1}{4}$.

- (a) Find and sketch the power spectral density of the output process.
- (b) Write down the 4×4 auto-correlation matrix for the output process.

4. [10 marks] A random process is given by $x[n] = a(u[n] - u[-n])$ where a is a uniform random variable in $[-1, 1]$ and $u[n]$ is the unit step signal.

- (a) Find the mean and auto-correlation functions for this process.
- (b) Is this process ergodic in the mean? That is, can the ensemble mean be estimated from any single realization? Justify.

5. [10 marks] A WSS random process $x[n]$ has the auto-correlation function given by the expression $r_x[k] = \alpha^{|k|} + \beta \delta[k]$. We want to model this process as an AR(2) process, i.e. an auto regressive process of order 2. Write down the Yule-Walker equations to estimate the parameters of this AR process and solve them in terms of α and β .